Smart Digital Measuring Tape Using ESP32 Supermini

Introduction

In the modern era of digital innovation, measuring tools have evolved from conventional tapes to smart devices that offer precision and ease of use. This project introduces a Smart Digital Measuring Tape powered by ESP32 Supermini, featuring a compact design, rotary encoder for accurate distance measurement, OLED display for real-time reading, and additional functionalities such as unit conversion (cm, m, ft) and battery level indication. The device aims to provide high accuracy, portability, and user-friendly operation for DIY, industrial, and construction applications.

Block Diagram

Rotary Encoder ---> ESP32 Supermini ---> OLED Display
↑ ↑
Reset Button Battery
Unit Button Power Source

Working Principle

The rotary encoder detects the wheel rotation and sends pulses to the ESP32 Supermini. Each pulse corresponds to a certain linear distance based on the wheel circumference. The ESP32 calculates the total distance traveled, converts it into selected units (cm, m, ft), and displays the measurement on the OLED screen. Additional buttons allow the user to reset the measurement or switch between units. A battery icon on the display shows the remaining power level for easy monitoring.

Supervisor

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